

LVA Ticket System — Design

Status: DESIGN (scaffold only; no DB binary, no generator scripts built yet) **Key prefix:** LVA (operator directive, §6.L 68th invocation, 2026-05-31) **Author:** design+scaffold subagent **Constitutional basis:** HelixConstitution §11.4.15 / §11.4.16 / §11.4.19 / §11.4.21 / §11.4.33 / §11.4.34 / §11.4.54 / §11.4.55 / §11.4.65; Lava §6.AD.3, §6.O, §6.T

0. The key decision

The operator (§6.L 68th invocation, 2026-05-31) directed: define the ticket key for the main workable-items SQLite database and related documents (Issues, Fixed, Issues_Summary, Fixed_Summary) and their exports to PDF / HTML / DOCX. **The key prefix is LVA.** Tickets are LVA-1, LVA-2, ... — a stable, monotonic, project-prefixed key. This is the Lava-specific instantiation of the HelixConstitution §11.4.54 ATM-NNN ticket-identifier mandate (ATM → LVA).

The seq INTEGER PRIMARY KEY AUTOINCREMENT column is the monotonic integer; the id column renders 'LVA-' || seq via an AFTER INSERT trigger. AUTOINCREMENT (not plain rowid) guarantees keys are never reused even after deletes — satisfying §11.4.54's "monotonic, never renumbered, append-only".

1. What exists today vs. what is missing

Thing	State (CONFIRMED via filesystem + sqlite3 + reading)
tickets.db / Issues.md / Fixed.md / Issues_Summary.md / Fixed_Summary.md	MISSING — none exist in the Lava repo (find returned no matches outside the HelixQA submodule + .claude/worktrees/)
docs/tickets/	CREATED by this scaffold (schema.sql + this DESIGN.md)
lava-api-go/internal/qa/ticket/*.go	IMPLEMENTED + REAL — generator.go (398 lines) + generator_test.go (426 lines). Adapter over HelixQA pkg/ticket.Generator that emits §6.O Crashlytics CLOSURE LOGS (one md per closed Crashlytics issue). It is NOT a tickets DB: no SQLite, no Issues/Fixed tracker, no LVA key. Different concern; see §6.
HelixQA submodule pkg/ticket (15 Go files)	ON DISK + REAL at submodules/helixqa/pkg/ticket/ (ticket.go 13.7 KB, enhanced_generator.go, rich_capture.go,

Thing	State (CONFIRMED via filesystem + sqlite3 + reading)
	OCU mapping/replay + tests). Emits HQA-#### generic QA-session tickets. NOT keyed LVA, NOT SQLite. The Lava adapter wraps this.
constitution/ submodule	FULLY CHECKED OUT — Constitution.md (7302 lines), CLAUDE.md (2485 lines) + pre-rendered .html/.pdf/.docx for each. Full §11.4.x text read for §2.
Equivalence-mapped state (§6.AD.3 Path B)	EXISTS — docs/CONTINUATION.md (§6.S) + .lava-ci-evidence/crashlytics-resolved/<date>-<slug>.md (§6.O) + .lava-ci-evidence/sixth-law-incidents/<date>-<slug>.json

Conclusion: no SQLite ticket DB nor Issues/Fixed/Summary tracker docs exist keyed LVA. The two existing Go ticket packages are a DIFFERENT artifact — Crashlytics/QA closure-log generators, not a workable-items database. This is a greenfield build keyed LVA that complements them.

2. HelixConstitution mandates this design satisfies

Quoted from the on-disk constitution/Constitution.md (7302 lines) and constitution/CLAUDE.md (2485 lines), both fully checked out and read:

- **§11.4.15 — Item-status tracking** (CLAUDE.md): *"Every active item in the project Issues file carries a **Status:** line... Six-state vocabulary: Queued, In progress, Ready for testing, In testing, Reopened, Fixed (→ Fixed.md)." §11.4.21 adds a 7th value Operator-blocked. → tickets.status FK to item_status. NOTE: the constitution's literal vocabulary differs from this design's normalized lifecycle states — the generator MUST render constitution-literal Status strings in the produced Issues.md (mapping in §4); the DB stores normalized states for query sanity.*
- **§11.4.16 — Item-type tracking** (CLAUDE.md): *"Three-value CLOSED vocabulary: Bug... Feature... Task..." → tickets.type FK to item_type. This design adds Chore/Incident/Debt as Lava-domain extras; the generator MUST collapse them to the constitution 3-value closed set in rendered docs (Chore→Task, Incident→Bug, Debt→Task) OR the operator ratifies an extension. CONFIRMED constitution closed set = {Bug, Feature, Task}.*
- **§11.4.19 — Fixed-document column-alignment** (Constitution.md line 1150+) + **§11.4.53 — Fixed_Summary**
parity: Fixed.md + Fixed_Summary.md MUST carry Status + Type columns aligned with Issues_Summary, and mtime(Fixed_Summary) ≥ mtime(Fixed). → v_fixed / v_fixed_summary VIEWS (in sync by construction); the generator emits all four .md + §11.4.65 .html + .pdf siblings.
- **§11.4.21 / §11.4.54 — stable project-prefixed key.** §11.4.54 (ATM-NNN mandate): *"Every workable item... MUST carry a stable, unique, auto-incremental ATM-NNN ticket identifier. ...monotonic, never renumbered, append-only."* For Lava the prefix is **LVA**. → the LVA-<seq> id with AUTOINCREMENT. §11.4.54 also requires the LVA-ID as leftmost column in both summaries.

- **§11.4.33 — Type-aware closure-status vocabulary** (CLAUDE.md): closed map Bug → Fixed (→ Fixed.md), Feature → Implemented (→ Fixed.md), Task → Completed (→ Fixed.md). → closure_status column + closure_status_for_type mapping + trg_closure_status_typeaware trigger (VALIDATED: Bug→Implemented rejected; Bug→Fixed accepted). The generator MUST emit the literal (→ Fixed.md) suffix.
- **§11.4.34 — Reopened-source attribution** (CLAUDE.md): every Reopened heading carries ****Reopened-Details:**** with **By** (AI|User), **On** (ISO), **Reason** (closed vocab {test-failed | manual-testing-detected | captured-evidence-contradicts | end-user-report | cycle-re-discovered | design-reconsidered}), **Evidence** (path). → reopened_who (By), reopened_when (On), reopened_why (Reason), reopened_incident (Evidence) columns + trg_reopen_attribution trigger (VALIDATED: reopen missing any of the four rejected). §11.4.55 adds per-item docs/issues/<LVA-NNN>/Reopens.md when reopens_count > 0 — see §7 (a ticket_events history table is the recommended follow-up).

Also honored: **§11.4.21 operator-blocked-details** → operator_blocked_details

- trigger; **§11.4.65 / §11.4.60** universal Markdown export (.md + .html + .pdf always in sync) → drives the export pipeline in §5.

Validation evidence: the schema was applied to a throwaway db with the host sqlite3 and every trigger was exercised — Bug→Implemented rejected (§11.4.33: Closed ticket needs a type-valid closure_status), Bug→Fixed accepted, reopen-without-attribution rejected (§11.4.34: Reopen needs reopened_why/who/when/incident), reopen-with-attribution accepted, LVA-1 / LVA-2 auto-rendered, closed_at auto-stamped, summary views populated. (No db committed — it is a generated artifact.)

3. Schema

Full DDL in docs/tickets/schema.sql. Shape:

- item_type, item_status, closure_status — closed-set lookup tables (FK-enforced; auditable + extensible without code edits).
- closure_status_for_type — the §11.4.33 (type → valid closure verbs) mapping.
- tickets — the core table (seq/id, title, body, type, status, closure_status, priority, source/source_ref, reopened_*, fix_commit_sha / validation_test / challenge_test / closure_log, operator_blocked_details, duplicate_of, timestamps).
- Triggers — id rendering, updated_at touch, type-aware closure guard, reopen attribution guard, operator-blocked guard, closed_at stamp.
- Views — v_issues, v_fixed, v_issues_summary, v_fixed_summary.

No sudo, no service, no network — host sqlite3 only.

4. Generated document set + vocabulary mapping

Four canonical markdown docs, generated FROM the db (single source of truth = the db; the .md files + their .html/.pdf/.docx are derived artifacts):

Doc	Source view	Content
Issues.md	v_issues	Non-closed tickets, leftmost LVA ID column (§11.4.54), then Type, Status, Priority, Title, Source
Fixed.md	v_fixed	Closed tickets, LVA ID, Type, Closure Status (literal (→ Fixed.md) per §11.4.33), Title, Fix SHA, Validation Test, Challenge Test, Closure Log
Issues_Summary.md	v_issues_summary	Counts by (Type × Status); §11.4.54 LVA-ID column on the detail rows
Fixed_Summary.md	v_fixed_summary	Counts by (Type × Closure Status); §11.4.19 column-aligned with Issues_Summary

DB-normalized → constitution-literal mapping the generator MUST apply:

DB status	Rendered **Status:**
Open	Queued
InProgress	In progress
InReview	Ready for testing / In testing
Reopened	Reopened (+ **Reopened-Details:** block per §11.4.34)
OperatorBlocked	Operator-blocked (+ **Operator-Block-Details:** per §11.4.21)
Closed	the type-aware closure verb (next table)

DB type + closure_status	Rendered closure **Status:** (§11.4.33)
Bug + Fixed	Fixed (→ Fixed.md)
Feature + Implemented	Implemented (→ Fixed.md)
Task + Completed	Completed (→ Fixed.md)

DB-extra types (Chore/Incident/Debt) collapse to {Task,Bug,Task} in rendered docs until the operator ratifies extending the constitution's 3-value set.

5. Export pipeline (PDF / HTML / DOCX) — built around what is ACTUALLY installed

Tool reality on this host (CONFIRMED — clean per-tool which + per-module p
ython3 -c "import X" probe; everything needed is present):

Tool	Result	Evidence
sqlite3	PRESENT	.../Android/sdk/platform-tools/ sqlite3 — schema + all triggers validated against it

Tool	Result	Evidence
python3	PRESENT	/opt/homebrew/opt/python@3.9/libexec/bin/python3
pandoc	PRESENT	/opt/homebrew/bin/pandoc — covers md→HTML AND md→DOCX
weasyprint	PRESENT (binary + py module 66.0)	/opt/homebrew/bin/weasyprint + import weasyprint OK — covers HTML→PDF
python python-docx (docx)	PRESENT (1.2.0)	import docx OK — programmatic DOCX fallback
python jinja2	PRESENT (3.1.6)	import jinja2 OK — optional templating
go	PRESENT	/opt/homebrew/bin/go — the Lava internal/qa/ticket adapter builds here
wkhtmltopdf	MISSING	not needed — weasyprint covers PDF
libreoffice / soffice	MISSING	not needed — pandoc covers DOCX
python markdown	MISSING	not needed — pandoc parses md

Earlier-probe correction (no-guessing, §11.4.6). A first probe printed a confusing interleaving (a stray Traceback aligned next to a later module's OK). The clean per-line probe (each import on its own invocation) is authoritative: pandoc + weasyprint + python-docx are ALL installed. The constitution submodule's pre-rendered .html/.pdf/.docx artifacts on disk corroborate that this exact pandoc + weasyprint pipeline already runs in this repo for §11.4.65 exports.

Implication: the FULL pipeline (md + HTML + PDF + DOCX) is buildable TODAY with no installs and no sudo. §11.4.65 / §11.4.60 (every tracked .md has synced .html + .pdf siblings; operator also asked for DOCX) is satisfiable now.

Export pipeline (single tier — all tools present):

1. **md generation:** sqlite3 query → Python (stdlib sqlite3; jinja2 optional) → Issues.md / Fixed.md / Issues_Summary.md / Fixed_Summary.md.
2. **HTML:** pandoc <doc>.md -o <doc>.html --standalone (matches the existing repo pattern that produced constitution/*.html).
3. **PDF:** pandoc <doc>.md -o <doc>.pdf --pdf-engine=weasyprint (the §11.4.65 documented "pandoc HTML + weasyprint PDF" combo).
4. **DOCX:** pandoc <doc>.md -o <doc>.docx (operator's explicit ask; pandoc native DOCX writer — no LibreOffice needed). python-docx is the fine-grained-styling fallback.

Mirror the constitution's sync_all_markdown_exports.sh timeout 60-per-file guard for consistency (§6.T.2 host-budget — these are sub-second conversions on a handful of small docs).

Operator-install items (NOT via sudo): NONE required for the core md/HTML/PDF/DOCX pipeline — every tool is already present. Optional future hardening: run pandoc+weasyprint inside the Containers submodule (§6.K / §6.U / §11.4.76) for cross-workstation reproducibility — not a blocker.

6. Relationship to the existing Go ticket packages + the §6.AD.3 mapping

The two existing ticket Go packages are a DIFFERENT artifact and are NOT superseded by the LVA system:

- submodules/helixqa/pkg/ticket/ (15 real Go files, ticket.go 13.7 KB) — HelixQA's Generator producing HQA-#### QA-session tickets (generic markdown for any QA failure: crash, step failure, ANR). Reusable, project-agnostic.
- lava-api-go/internal/qa/ticket/generator.go (398 real lines) — Lava's adapter wrapping the HelixQA Generator to emit §6.O Crashlytics **closure logs** at .lava-ci-evidence/crashlytics-resolved/<date>-<slug>.md. One file per closed Crashlytics issue.

Neither manages an Issues/Fixed workable-items tracker, neither uses SQLite, neither is keyed LVA. They are output generators for incident/QA artifacts. The LVA system is the **workable-items database** (the operator's directive target). They **COMPLEMENT** each other: a Crashlytics issue closed via the internal/qa/ticket adapter **SHOULD** also become an Incident/Bug LVA ticket (status Closed, closure_log pointing at the generated md). Per §11.4.74 (catalogue-first / extend-don't-reimplement): the LVA generator **MAY** reuse HelixQA pkg/ticket rendering helpers where they fit rather than re-rolling markdown — flagged in §7. The Go adapter is **REAL** and working — do **NOT** delete it.

Relationship to the §6.AD.3 Path-B equivalence mapping

§6.AD.3 Path B (2026-05-15) formally mapped HelixConstitution §11.4.33 / §11.4.34 to Lava's existing artifacts (CONTINUATION.md + the two .lava-ci-evidence/ ledgers) and declared that mapping binding "rather than requiring a parallel Issues.md / Fixed.md tracker."

This LVA system COMPLEMENTS rather than immediately supersedes that mapping:

- The operator's 68th-invocation directive now explicitly calls for the SQLite DB
 - Issues/Fixed/Summary docs + PDF/HTML/DOCX exports — i.e. the very parallel tracker Path B previously deemed optional. The LVA system is the operator-directed materialization of those trackers.
- Until the LVA system is populated and wired into the gates, the Path-B equivalence remains the binding compliance surface (do **NOT** remove the .lava-ci-evidence/ ledgers or CONTINUATION.md state). The LVA db should be **SEEDED** from the existing .lava-ci-evidence/crashlytics-resolved/* + sixth-law-incidents/* so the two surfaces stay reconciled.
- Once the LVA db is authoritative and the generators run in CI (scripts/ci.sh), §6.AD.3 should be amended to note the LVA system as primary tracker

with the `.lava-ci-evidence/` ledgers as forensic backing (a `CLAUDE.md` edit for the main agent / operator — NOT done in this scaffold).

This supersede-vs-complement decision is project-specific (`Classification: project-specific`) and must be ratified by the operator before the Path-B mapping text is rewritten.

7. Action list for the main agent (prioritized)

1. **Commit this scaffold** (`docs/tickets/DESIGN.md` + `docs/tickets/schema.sql`). Per §6.S update `docs/CONTINUATION.md` in the same commit (new `docs/tickets` surface + LVA key decision). Add a `Classification:` line per §11.4.17.
 2. **Build the generator** — `scripts/tickets-gen.py` (`stdlib sqlite3`; `jinja2` optional, present) reads the db and emits `Issues.md` / `Fixed.md` / `Issues_Summary.md` / `Fixed_Summary.md`, then shells to `pandoc` (HTML + DOCX) + `pandoc --pdf-engine=weasyprint` (PDF) per §5 — ALL tools confirmed present, no install. Add a `docs/scripts/tickets-gen.md` companion per §11.4.18. Cover with a hermetic test per §6.A. Apply the §4 normalized→constitution-literal vocab mapping so rendered docs satisfy §11.4.15/16/33.
 3. **Seed from existing evidence** — import `.lava-ci-evidence/crashlytics-resolved/*` (→ Bug/Incident, Closed, `closure_log` set) and `sixth-law-incidents/*` (→ Incident) as LVA tickets, reconciling with the Path-B surface (§6).
 4. **§11.4.74 catalogue-check before re-rolling markdown.** Evaluate reusing/ extending the HelixQA `pkg/ticket` + Lava `internal/qa/ticket` rendering helpers; record `Catalogue-Check: reuse|extend|no-match`. Do NOT delete the Go adapter (different concern — closure logs, not the tracker DB).
 5. **Decide canonical tracker-doc location.** §11.4.48 carve-out fixes the 5 canonical tracker docs at `docs/ ROOT` (`docs/Issues.md`, ..., `docs/CONTINUATION.md`). If Lava adopts that convention, rendered tracker docs go at `docs/ root` with the SQLite db + schema + generator under `docs/tickets/`. Operator decision via §11.4.66 `AskUserQuestion`.
 6. **Gitignore decision for tickets.db.** Recommend gitignore the binary + commit `schema.sql` + a `seed.sql` of INSERTs (reproducible, reviewable), with a §11.4.77 `.gitignore-meta/` entry declaring `sqlite3 db < seed.sql` as the regeneration mechanism. (Existing precedent: `.codegraph/ codegraph.db` is gitignored with `codegraph index` as its regen mechanism.)
 7. **No operator install required** for md/HTML/PDF/DOCX — `pandoc` + `weasyprint` + `python-docx` + `sqlite3` + `python3` + `go` all present (evidence §5).
 8. **Wire gates** — add a `scripts/check-constitution.sh` check that the LVA db's triggers exist + generated docs are in sync (`regenerate + git diff --exit-code`) + `.md/.html/.pdf` mtimes aligned (§11.4.60/65), closing the §6.AD-debt CM-ITEM-* gates with a real implementation instead of equivalence-mapping. Add a `Classification:` line + paired §1.1 mutation test.
 9. **Follow-up: ticket_events history table** for full §11.4.55 per-item `Reopens.md` reasoning chains (the current `reopened_*` columns hold only the latest reopen; full history needs an append-only events table).
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8. Build outcome (2026-05-31, §6.L 68th invocation)

The system is BUILT and the design's open questions are resolved. Verbatim build evidence: docs/tickets/BUILD-EVIDENCE.md.

- **Implementation language: Go, not Python.** The §7.2 design sketched a scripts/tickets-gen.py; the actual build is a Go module at tools/lava-tickets/ (pure modernc.org/sqlite, no CGO) per the operator's Go-only directive. It implements init / add / update / close / reopen / gen / verify / import / export / list.
- **DB-in-git decision (§7.6 superseded):** docs/tickets/tickets.db IS **TRACKED in git, NOT gitignored** — §11.4.95 ("Workable-items SQLite DB TRACKED in git, NEVER gitignored", a NEW clause in the 53-commit delta the 68th-cycle review surfaced) is explicit and overrides §7.6's gitignore-the-binary recommendation. The DB is checkpointed to DELETE journal mode before commit so it is self-contained; only *.db-wal/*.db-shm sidecars + bin/ + scratch_*.sh are gitignored.
- **Round-trip (§11.4.106):** gen then verify is byte-identical — PROVEN, and the contract test includes a falsifiability rehearsal (corrupting a tracker makes verify FAIL).
- **Triggers (§11.4.33/34):** exercised by the binary's contract test — Bug→Implemented rejected, Bug→Fixed accepted, partial reopen rejected.
- **Export:** HTML is pure-Go (always works — Issues.html committed as proof). PDF/DOCX run pandoc inside a podman/docker container; when no runtime is up the tool exits 3 with an operator-actionable message and writes NO fake file (§6.J). Host pandoc is the documented operator fallback (real DOCX produced this session).
- **Seeded items:** 7 REAL items (LVA-1...LVA-7) derived from docs/CONTINUATION.md
 - the 68th-cycle constitution review + the two 2026-05-20 sixth-law incidents. No invented commit SHA (recorded as pending per §11.4.6 no-guessing).
- **§6.AD.3 Path-B vs SQLite-DB reconciliation (LVA-3) remains operator-gated** — this build materializes the tracker the §6.AD.3 carve-out previously deemed optional; whether it SUPERSEDES or COMPLEMENTS the .lava-ci-evidence/ ledgers is the main-agent/operator decision (DESIGN §6), not made in this build.

Classification: project-specific (the LVA key + Lava's doc set + the Path-B relationship are Lava-specific; the underlying §11.4.x mandates are universal and inherited from HelixConstitution).